

Exact transmission conditions and inclusions for finite-step variable bandwidth

Automated mathematical discovery run `fa_banach_001`, lane 11

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Status. Likely valid full answers to Questions II.5 and II.6 of arXiv:2503.22443, plus a sharp one-sided result toward the sampled-sign question.

Source questions and normalization

Jaming and Perez ask when local Paley–Wiener pieces patch to their finite-step variable-bandwidth space, and whether an order between two step coefficients implies inclusion of the corresponding spaces. The displayed comparison in Question II.6 reads $p_1 \geq p_2$ although the two functions are called $p, \tilde{p}$; below it is read in the natural intended sense $p \geq \tilde{p}$ pointwise.

Corollary II.4. *Let $p \in PC_N$, A_p be the self-adjoint realization of τ_p and $\Omega > 0$. Then $PW_\Omega(A_p)$ is a closed subspace of $L^2(\mathbb{R})$ consisting of continuous functions. Moreover, on each I_j , f is the restriction to I_j of a function in $PW_{\Omega q_j}(\mathbb{R})$.*

The main difference is that in [3], the authors prove that f is locally in the larger Bernstein space. On the other hand, their result is valid for more general p . This corollary also leads to the following question:

Question II.5. *Let $p \in PC_N$ and, for each j , let f_j be the restriction to I_j of a function in $PW_{\Omega q_j}(\mathbb{R})$. Under which*

condition is the function $f = \sum_{j=0}^N f_j \chi_{I_j} \in PW_\Omega(A_p)$?

Of course, f has to be continuous, but this is not enough. So far, we have only been able to answer this question in the single jump case. This will be done in the next section and also leads to another proof of the sign retrieval question in this case. It is likely that this would also be a crucial step for phase retrieval in variable band width spaces.

This raises also a second question:

Question II.6. *Let $p, \tilde{p} \in PC_N$ be such that $p_1 \geq p_2$. Is $PW_\Omega(A_p) \subset PW_\Omega(A_{\tilde{p}})$?*

From the formula, this is clear when $\tilde{p} = \alpha p$, but not in the general case.

It is cleaner to separate the spectral cutoff from the paper's competing Ω normalizations. Let

$$A_p f = -(pf')', \quad \mathcal{V}_p(B) = \text{Ran } \mathbf{1}_{[0,B]}(A_p), \quad B > 0.$$

Suppose $p = p_j > 0$ on $I_j = (x_j, x_{j+1})$, where $-\infty = x_0 < x_1 < \cdots < x_N < x_{N+1} = \infty$, and put

$$c_j = \sqrt{B/p_j}.$$

Thus an element of $\mathcal{V}_p(B)$ restricts on I_j to an entire member of the classical $PW_{c_j}(\mathbb{R})$. In the explicit formulas of the source, $c_j = \Omega p_j^{-1/2}$, so $B = \Omega^2$.

Full answer to the local-patching question

For each j , let $F_j \in PW_{c_j}(\mathbb{R})$, and define $f(x) = F_j(x)$ for $x \in I_j$. Values and derivatives below are those of the entire functions F_j .

Theorem 1 (Transmission-jet characterization). *The piecewise function f belongs to $\mathcal{V}_p(B)$ if and only if, at every jump x_j and for every $m \geq 0$,*

$$p_{j-1}^m F_{j-1}^{(2m)}(x_j) = p_j^m F_j^{(2m)}(x_j), \quad (\text{T}_{2m})$$

$$p_{j-1}^{m+1} F_{j-1}^{(2m+1)}(x_j) = p_j^{m+1} F_j^{(2m+1)}(x_j). \quad (\text{T}_{2m+1})$$

In particular, continuity is only the first of an infinite, explicit family of matching conditions.

Proof. If $f \in \mathcal{V}_p(B)$, then $f \in \text{Dom}(A_p^m)$ for every m . A function u in $\text{Dom}(A_p)$ is continuous at every jump and has continuous flux pu' there. On I_j ,

$$A_p^m f = (-1)^m p_j^m F_j^{(2m)}. \quad (1)$$

Applying continuity and flux continuity to $A_p^m f$ gives (T_{2m}) and (T_{2m+1}) .

Conversely, the conditions with $m = 0$ put f in $\text{Dom}(A_p)$. The complete family, used inductively, puts f in every $\text{Dom}(A_p^m)$ and proves (1). Bernstein's inequality in PW_{c_j} now gives, with $C^2 = \sum_j \|F_j\|_2^2$,

$$\begin{aligned} \|A_p^m f\|_2^2 &\leq \sum_j p_j^{2m} \|F_j^{(2m)}\|_2^2 \leq \sum_j p_j^{2m} c_j^{4m} \|F_j\|_2^2 \\ &= C^2 B^{2m}. \end{aligned} \quad (2)$$

Let μ_f be the spectral measure of f for A_p . If $\mu_f((B + \varepsilon, \infty)) = \delta > 0$, then the spectral theorem gives

$$\|A_p^m f\|_2^2 = \int \lambda^{2m} d\mu_f(\lambda) \geq \delta (B + \varepsilon)^{2m},$$

contradicting (2) as $m \rightarrow \infty$. Hence $\text{supp } \mu_f \subset [0, B]$ and $f \in \mathcal{V}_p(B)$. $\square$

The conditions also reveal why one local piece determines the global function. At a jump x , write the adjacent constants as p_-, p_+ and set $H_-(t) = F_-(x + \sqrt{p_-} t)$. Then the unique compatible right piece is given by

$$\begin{aligned} H_+(t) := F_+(x + \sqrt{p_+} t) &= \frac{H_-(t) + H_-(-t)}{2} \\ &\quad + \sqrt{\frac{p_-}{p_+}} \frac{H_-(t) - H_-(-t)}{2}. \end{aligned} \quad (3)$$

Indeed, the even Taylor coefficients in (3) encode (T_{2m}) , and the odd ones encode (T_{2m+1}) . Both sides have exponential type at most $\sqrt{B}$ in t , and reflection and dilation preserve L^2 membership. Iteration and Theorem 1 yield the following useful strengthening of the source's uniqueness observation.

Corollary 1 (Onto local chart). *For every component I_j , restriction followed by entire continuation is a real-linear bijection*

$$\mathcal{V}_p(B)_{\mathbb{R}} \longrightarrow PW_{\sqrt{B/p_j}}(\mathbb{R})_{\mathbb{R}}.$$

Full answer to the inclusion question

The next theorem allows the finite jump sets of p and $\tilde{p}$ to be different.

Theorem 2 (Complete inclusion classification). *For positive finite-step functions $p, \tilde{p}$ and $B > 0$,*

$$\mathcal{V}_p(B) \subseteq \mathcal{V}_{\tilde{p}}(B) \iff \tilde{p} = \alpha p \text{ for some } 0 < \alpha \leq 1. \quad (4)$$

Consequently, even under the pointwise order $p \geq \tilde{p}$, inclusion fails unless the coefficients are global scalar multiples.

Proof. Assume the inclusion. At any breakpoint x in the union of the two step partitions, denote the one-sided constants by $p_{\pm}, \tilde{p}_{\pm}$; the two constants are equal if the corresponding function has no jump there. Every $f \in \mathcal{V}_p(B)$ must satisfy both flux laws

$$p_- f'(x-) = p_+ f'(x+), \quad \tilde{p}_- f'(x-) = \tilde{p}_+ f'(x+). \quad (5)$$

If $p_-/p_+ \neq \tilde{p}_-/\tilde{p}_+$, the two equations force both derivatives to vanish for every $f \in \mathcal{V}_p(B)$. This is impossible by Corollary 1: in either adjacent local chart, prescribe a real Paley–Wiener function whose derivative at x is nonzero. Thus the ratios agree at every breakpoint. The quotient $\tilde{p}/p$ is consequently constant across the common refinement, say $\tilde{p} = \alpha p$.

Conversely, $A_{\alpha p} = \alpha A_p$ on the same domain, and therefore

$$\mathcal{V}_{\alpha p}(B) = \mathcal{V}_p(B/\alpha). \quad (6)$$

Equation (6) gives the inclusion exactly when $\alpha \leq 1$. It is strict in the opposite direction: a local chart contains functions with Fourier mass between the two distinct local band edges. $\square$

Here is a concrete negative answer to the proposed monotonicity. With one jump at zero, take

$$p(x) = \begin{cases} 4, & x < 0, \\ 1, & x > 0, \end{cases} \quad \tilde{p}(x) = \begin{cases} 2, & x < 0, \\ 1, & x > 0. \end{cases} \quad (7)$$

Then $p \geq \tilde{p}$. In the left local chart let $c = \sqrt{B}/2$ and prescribe

$$\hat{F}(\xi) = -i\xi \mathbf{1}_{[-c, c]}(\xi).$$

This defines a real $F \in PW_c$ with $F'(0) = 2c^3/(3\sqrt{2\pi}) > 0$. Its lift lies in $\mathcal{V}_p(B)$. If it also lay in $\mathcal{V}_{\tilde{p}}(B)$, (5) would simultaneously give $f'(0+) = 4f'(0-)$ and $f'(0+) = 2f'(0-)$, a contradiction.

Sharp tail sampling and an all-bandwidth set

The source's final question asks for discrete sign-retrieval sets. The full multi-component classification remains open, but the local chart gives a sharp answer for a natural family.

Theorem 3 (Sharp one-sided lattice). *Fix the right unbounded component $I_N = (x_N, \infty)$, choose $a > x_N$, and put $c = \sqrt{B/p_N}$. The one-sided lattice*

$$\Lambda_h = \{a + nh : n \geq 0\}$$

sign retrieves every real member of $\mathcal{V}_p(B)$ if and only if

$$0 < h < \frac{\pi}{2c}. \quad (8)$$

Proof. By Corollary 1, it suffices to work in real PW_c . Suppose $h < \pi/(2c)$ and $|F| = |G|$ on Λ_h . Then $H = F^2 - G^2 \in PW_{2c}$ and $H(a + nh) = 0$ for every $n \geq 0$. If $\varphi(\xi) = e^{ia\xi} \widehat{H}(\xi)$, these zeros say that φ is orthogonal on $[-2c, 2c]$ to every nonnegative mode $e^{inh\xi}$. After the change of variable $\theta = h\xi$, extend $\varphi(\theta/h)$ by zero to the circle. Its support is a proper arc because $4ch < 2\pi$, while the mode conditions put it in one Hardy boundary subspace. Hardy boundary uniqueness forces it to vanish: a nonzero Hardy function cannot have boundary values zero on the complementary arc of positive measure. Thus $H = 0$. Since entire functions form an integral domain, $(F - G)(F + G) = 0$ implies $F = G$ or $F = -G$.

For sharpness, let $h \geq \pi/(2c)$ and $d = \pi/(2h) \leq c$. Fill the removable singularities in

$$U(z) = \frac{\sin(d(z - a))}{z - a + 2h}, \quad V(z) = \frac{\cos(d(z - a))}{z - a + h}. \quad (9)$$

These are nonzero real members of PW_c : they have type d and are square integrable on $\mathbb{R}$. At even samples $U = 0$, and at odd samples $V = 0$. Hence $F = U + V$ and $G = U - V$ have the same sampled magnitudes but are not global signs of one another. Their local-chart lifts prove failure in $\mathcal{V}_p(B)$. $\square$

There is also one discrete set that works for every finite bandwidth at once. For $a > x_N$, let

$$\Lambda_\infty = \{a + \sqrt{n} : n \geq 1\}. \quad (10)$$

If a nonzero entire function has finite exponential type, Jensen's formula gives only $O(r)$ zeros in $|z| \leq r$, whereas (10) contributes order r^2 . Thus (10) is a uniqueness set for PW_{2c} for every finite c . Applying the squared-function argument above proves simultaneous sign retrieval in $\mathcal{V}_p(B)_\mathbb{R}$ for every $B > 0$. This resolves the literal bandwidth-free reading of the source's notation $PW(A_p)$, though the set is not separated and no stability claim is made.

Scope, novelty, and human-review focus

Theorems 1 and 2 fully answer the two questions reproduced in the crop. Theorem 3 and (10) answer substantial one-component cases of the final sampled-sign question, not its arbitrary multi-component classification.

A bounded search through 17 August 2026 used the exact source title and questions together with “variable bandwidth inclusion,” “step Sturm–Liouville spectral subspace,” and “sampled sign retrieval.” It found the source and Thakur's classical unsigned Paley–Wiener theorem, but no answer to these finite-step questions. Novelty is therefore only “apparently new within the bounded search.”

Human review should focus on (i) the source's inconsistent Ω cutoff, which is why the packet uses the unambiguous edge B ; (ii) the induction from the jet conditions to $\text{Dom}(A_p^m)$; and (iii) the intended reading of the type in Question II.6. None affects the operator-theoretic statement in the B normalization.

References

1. P. Jaming and R. Perez III, “Sign retrieval in spaces of variable bandwidth,” arXiv:2503.22443 (2025), Questions II.5, II.6, and Section IV.
2. K. Gröchenig and A. Klotz, “What is variable bandwidth?” *Comm. Pure Appl. Math.* 70 (2017), 2039–2083.
3. G. Thakur, “Reconstruction of bandlimited functions from unsigned samples,” *J. Fourier Anal. Appl.* 17 (2011), 720–732; arXiv:1007.0198.